IV / IO Access

UP-06 | Universal Patient Care

Quick flow

01 Is Vascular Access Needed? [EMT / AEMT / Paramedic]

Medication • fluid • time-critical treatment • anticipated deterioration

02 Immediate IO Indication? [AEMT / Paramedic]

Cardiac arrest • profound shock • critical treatment cannot wait

03 Peripheral IV / Saline Lock [AEMT / Paramedic]

Best conventional site • maximum 3 total attempts across the crew • do not delay transport

04 Critical Access Still Needed? [AEMT / Paramedic]

After unsuccessful peripheral attempts, proceed to IO

05 Preferred IO Site [AEMT / Paramedic]

Proximal tibia • adult humeral head only when tibial access is contraindicated

06 Confirm + Secure + Reassess [AEMT / Paramedic]

Patency • flush • secure • inspect for infiltration • document site and every attempt

Overview

- Establish vascular access only when needed for medication, fluid, blood sampling, or anticipated time-critical treatment. Use the least invasive route that meets the patient’s immediate clinical need.
- Do not delay transport or a time-critical medication solely to obtain peripheral IV access. In cardiac arrest, profound shock, or another immediately life-threatening condition, IO access may be established without preceding IV attempts.

Provider-Level Actions

Higher provider levels include the actions listed for the preceding levels.

EMT

- Identify whether the patient may require medication, fluid, blood sampling, or another time-critical treatment and notify the AEMT or Paramedic.
- Prepare and assist with vascular-access equipment, position the extremity, maintain aseptic technique, monitor the patient, and document assistance as appropriate. EMT personnel do not insert IV or IO catheters under this protocol.

AEMT

- Establish a peripheral IV or saline lock when clinically indicated. Use an eligible upper-extremity site; do not cannulate a leg or neck. Limit AEMT peripheral attempts to two in one arm. Select a conventional peripheral site appropriate for the treatment and avoid an injured extremity, an extremity with a dialysis fistula or graft, or a site with infection or impaired circulation.
- Limit peripheral IV attempts to three total attempts across the entire crew. Do not allow repeated attempts by successive providers to restart the count. If vascular access is critical after unsuccessful attempts, proceed to IO access.
- In cardiac arrest, profound shock, or another immediately life-threatening condition in which peripheral attempts would delay critical treatment, proceed directly to IO access without a required preceding IV attempt.

- Use the proximal tibia as the preferred IO site. In adults, the humeral head may be used as an alternative when tibial access is contraindicated and appropriate landmarks can be identified. Use proximal tibial access for pediatric patients younger than 16 years.
- For a conscious patient, administer 2% preservative-free, epinephrine-free lidocaine suitable for IV/IO administration slowly through the IO before the initial forceful flush: adult 20-40 mg; pediatric 0.5 mg/kg, maximum 40 mg. Flush and operate the device according to manufacturer instructions, using pressure-assisted infusion when needed.
- Confirm catheter stability and flow, secure the catheter, and inspect repeatedly for swelling, leakage, displacement, increasing pain, resistance, or impaired distal perfusion. Stop using the site immediately if infiltration or another complication is suspected.

Paramedic

- Perform the AEMT vascular-access actions and select access according to the medication, resuscitation, anticipated flow requirement, patient anatomy, and transport priority.
- When conventional peripheral access is unsuccessful or unavailable and IV access remains clinically necessary, establish an external-jugular IV. Do not use the external jugular as a routine first site and do not place a saline lock in an external-jugular vein.
- Do not access a PICC, central venous catheter, implanted port, or dialysis catheter under UP-06. Existing central-device use during interfacility transport is governed by the separate interfacility-transfer policy.

Indications

- Medication, fluid, or blood sampling required by the applicable treatment protocol.
- Critical illness, cardiac arrest, profound shock, or anticipated deterioration for which vascular access is needed.

Contraindications

- Do not place an IO through infection, into a fractured target bone, or at a site with a relevant prosthesis or prior significant orthopedic procedure. Follow the current instructions for the agency-stocked device. For Arrow EZ-IO, do not use a target bone that has had IO access or an attempted IO insertion within the previous 48 hours; select another eligible bone.
- Do not place an IO when landmarks cannot be identified or the device needle is not appropriate for the patient’s tissue depth.
- Do not place a peripheral IV in an injured extremity, an extremity with a dialysis fistula or graft, or a site with infection or severely impaired circulation when another site is available.

Assessment | Access decision

- Determine the treatment that requires access, the urgency of that treatment, expected flow needs, transport priority, and whether a nonvascular route can provide the required medication without delay.
- Before IO placement, inspect the proposed bone and overlying tissue for fracture, infection, orthopedic hardware or prosthesis, recent IO use, excessive tissue depth, and identifiable landmarks.

Treatment steps

- Use a peripheral IV or saline lock when access is needed and the patient’s condition permits the attempt.
- Limit peripheral IV attempts to three total attempts across the entire crew; proceed to IO when access remains critical.
- Proceed directly to IO in cardiac arrest, profound shock, or another immediately life-threatening condition when IV attempts would delay treatment.
- Use the proximal tibia as the preferred IO site. Use the adult humeral head only when tibial access is contraindicated; use the proximal tibia for pediatric patients.

- For a conscious IO patient, administer 2% preservative-free, epinephrine-free lidocaine suitable for IV/IO administration before flushing, then flush, secure, and monitor the site continuously.
- Document the indication, provider, site, catheter or needle size, number and location of every attempt, medications administered, patency, complications, and response.

Medications

Lidocaine 2% - conscious IO infusion pain

Adult: 20-40 mg IO administered slowly before the initial forceful flush. Pediatric: 0.5 mg/kg IO administered slowly before the initial forceful flush; maximum 40 mg.

- Use only 2% preservative-free, epinephrine-free lidocaine suitable for IV/IO administration. AEMT/Paramedic within device training and local authorization. This dose is for IO infusion pain, not dysrhythmia treatment.
- Confirm the IO is appropriately positioned and secure before medication administration.

Warnings / precautions

- AEMT peripheral IV access: use an eligible upper-extremity site; do not cannulate a leg or neck. Limit AEMT peripheral attempts to two in one arm. The maximum of three peripheral attempts across the entire crew is an overall cap, not permission to exceed a provider-specific limit or a requirement to delay indicated IO access.
- Three peripheral attempts is the maximum across the entire crew, not a requirement to perform three attempts before IO access.
- Do not delay transport, resuscitation, or a time-critical medication for repeated vascular-access attempts.
- IO infusion is painful in conscious patients. Administer lidocaine before the initial forceful flush unless an immediate life threat prevents delay.
- Stop using an IV or IO immediately for swelling, leakage, displacement, increasing pain, resistance, loss of flow, or suspected infiltration or compartment syndrome.
- Central venous devices and dialysis catheters are outside UP-06 and may be used only under the separate interfacility-transfer policy.

Clinical pearls

- A functioning IO provides access for resuscitation medications and fluids when peripheral access is not rapidly available, but pressure-assisted infusion is commonly required.
- Aspiration of marrow or blood may support placement but is not required when the catheter is stable, flushes appropriately, and there is no evidence of infiltration.
- For trauma patients, establish access during transport whenever feasible rather than extending scene time solely for an IV.

Special populations | Pediatrics - younger than 16 years

- Use the proximal tibia for IO access and select needle length according to patient size and manufacturer guidance.
- For a conscious child, administer lidocaine 0.5 mg/kg IO slowly before the initial forceful flush; maximum 40 mg.

Source and references

Tennessee EMS BLS/ALS State Protocol Guidelines 2024-2025 - Vascular Access, Intraosseous Access, and Indwelling IV Port Access procedures | Claiborne County EMS UP-06 source PDF - IV or IO Access, revised April 6, 2026 | Claiborne County EMS UP-01 Universal Patient Care and separate interfacility-transfer policy

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